

Care Creations™

BASF
We create chemistry

Irwinol®

The tropical solution for dry skin and lips

by Beauty Creations
another Care Creations™ product group Inspired by Life

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SUMMARY FILE

Irwinol®	LS 9890
Origin - Description	Botanical wax from *Irvingia gabonensis* mainly consisting of triglycerides of myristic and lauric acid
Regulatory data	
INCI	Octyldodecanol (and) Irvingia Gabonensis Kernel Butter (and) Hydrogenated Coco-Glycerides
China	Each component is listed in Inventory of Existing Cosmetic Ingredient in China (IECIC 2015)
CAS#	5333-42-6 ; 999999-99-4 (or 192230-28-7) ; 91744-42-2
EINECS#	226-242-9 ; 310-127-6 (n/a) ; 294-604-3
Appearance	White to pale yellow paste with a weak odor
Preservative	None
Natural labels	Raw material conforms to COSMOS standard of natural and organic cosmetics. This Product is RSPO-Mass Balance certified (BMT-RSPO-000099)
Naturalness content (ISO 16128)	100% from natural origin
Cosmetic use	
Properties	Moisturizing and softening efficacy on skin Moisturizing and repairing effect on lips
Applications	Moisturizing face and body care products Treatment products for hands and feet Ethnic skin and hair care Alternative to Shea butter Moisturizing and repairing lip care
Formulation data	
Concentration of use %	Skin Care: 1 - 3 / Make-Up: 0.5 - 3 / Hair Care: 0.5 - 1
Solubility	Soluble in oils and fats at T>40°C, insoluble in water
Incorporation method	Irwinol is dissolved in the fatty phase of emulsions at 80°C just before the emulsification process or melted at 40°C for incorporation in gels of foaming products.
Optimal pH	3 - 12

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A DELICIOUS EXOTIC BUTTER TO “TREAT” SKIN AND LIPS

Irwinol is a botanical wax, obtained by extraction from an African fruit called “Wild Mango”. It is proposed to exotically replace usual shea butter, for hydrating, nourishing and repairing properties of dry skin and dry lips.

Key properties:

- Moisturizing, soothing and softening effect on skin. Irwinol improves skin comfort and suppleness.
- Repairing activity on damaged dry lips. Their visible benefits were approved by consumers.
- Add functional benefits in facial make-up formula. The caring effects of the wild mango butter were also perceived by the consumers in a lip make-up formulation.

Irwinol is a botanical wax, obtained by extraction from an exotic fruit: Irvingia Aubry gabonensis, called “Wild Mango”

Description of the plant





Family: Irvingiaceae

Species: Irvingia gabonensis

Common name: Bush mango, African mango or Dika nut

Description: *Irvingia Aubry gabonensis* can reach up to 35 meters (100 feet) of height. The fruit has a diameter of 4 to 7 cm. It turns yellow at maturity and looks like a small mango.

Origin: Togo, Ivory Coast

Plant part used: Kernel

Distribution: it is found in the tropical and subtropical forests of west on central africa.

The “bush mango”



Irvingia gabonensis is an indigeneous wild tree native to the humid zones of West and Central Africa. It grows in the tropical and subtropical forests of Central and West Africa, attaining a height of up to 35 meters (100 feet). Its fruits are 4 to 7 cm in diameter and yellow at maturity. Although it does not belong to the same botanical family as the cultivated mango, this tree is commonly called wild mango tree or African mango tree because its fruit resembles a small mango. Due of its excellent nutritional value, the edible pulp is often used for preparing food dishes.

Traditional use of almonds

Food

- As a vegetable butter (solid oil): used in cooking and as cocoa butter substitute or margarine (Yamoneka *et al*, 2019).
- As a flour or thickner: used after grinding as a soup, sauce or stew additive, for improving the flavour and consistency (Njoku and Ugwuany, 1997).

Hygiene

- To make soaps.

Pharmaceutical uses

- As lubricant for tablets and suppositories (Udeala *et al*, 1980; Udeala and Salah, 2008; Megwa, 1987).
- Against metabolic disorders:
  - anti-intestinal enzymatic, anti-cholesterol and anti-hypoglycemic activities (Lieb *et al*, 2019; Omoruyi and Adamson, 1993; Omoruyi and Adamson, 1994),
  - inhibits *in vitro* adipogenesis to treat obesity and insulin resistance (Oben *et al*, 2008; Ngondi *et al*, 2009),
  - antimicrobial activities against *S. aureus* (Wandji *et al*, 2019).

Processing and composition

Irwinol is a botanical butter, extracted from the kernel of the edible fruit of an exotic tree, Irvingia gabonensis.

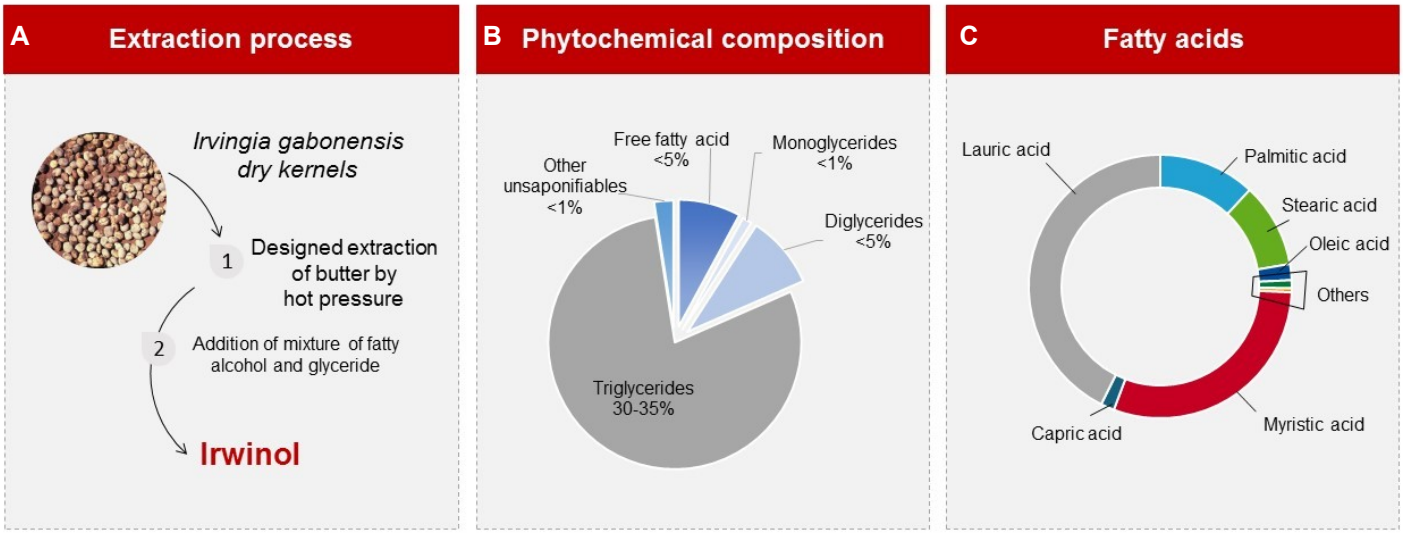


Figure 1 - Phytochemical data. **A** - Irwinol extraction process. **B** - Typical phytochemical composition of Irwinol (based on 2 industrial batches). **C** - Typical phytochemical composition of fatty acids of the lipid fraction (based on 2 industrial batches).

Safety / tolerability of the product

Irwinol was tested to ensure its safety in the recommended conditions of use. Irwinol does not irritate the eyes or skin and no indication of skin sensitization was observed.

Description

Irwinol contains a lipid extract from the kernels of Irvingia gabonensis, mainly consisting of triglycerides of myristic and lauric acid, with a minor unsaponifiable fraction of sterols and tocopherol.

Active fraction: Primarily triglycerides of myristic acid (40 - 59%) and lauric acid (35 - 52%), with a minor unsaponifiable fraction containing sterols and tocopherol.

Excipient: Mixture of fatty alcohol and glyceride.

INCI name: Octyldodecanol (and) Irvingia Gabonensis Kernel Butter (and) Hydrogenated Coco-Glycerides

Naturalness content (ISO 16128): 100% from natural origin

Dose of use: 0.5 - 3%



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Example: Illustration of a commercial sample of Irwinol - Emulsion and a lip lacquer at 3%.



DEMONSTRATED EFFICACY

Irwinol is a botanical butter having excellent natural lubricant, hydrating and softening properties.

Proven performance

Irwinol was evaluated through *ex vivo and in vivo* studies to demonstrate its benefits for skin care and lips applications:

- **moisturizing** and **softening** efficacy on skin. It protects and repairs skin through its film-forming activity, thereby restoring softness and suppleness. A clinical test has shown that Irwinol is a better skin moisturizer than shea butter,
- **moisturizing** and **repairing effect** on lips. It improves the appearance of dry and damaged lips *in vivo* by smoothing the scales. Interestingly, these functional benefits can be added to make-up formula. The caring effects of the wild mango butter were also perceived by the consumers in a make-up formulation for lips.

EFFICACY

Moisturizing activity *ex vivo*

ex vivo

OBJECTIVE

The moisturizing effect on the *stratum corneum* of Irwinol was compared to that of Shea butter and a placebo. The moisturizing activity was evaluated by measuring the dielectric conductivity of the *stratum corneum*, which is proportional to its moisture content.

PROTOCOL

A sample of *stratum corneum* is isolated from skin.
Its moisture level is measured by determining its dielectric conductivity:
- before treatment,
- after treatment (3 times at 30 min intervals) with:
  . a placebo cream,
  . a cream containing 2% Shea butter,
  . a cream containing 2% Irwinol.

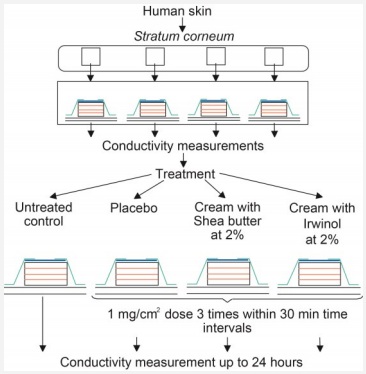


Figure 2 - Protocol.

RESULTS

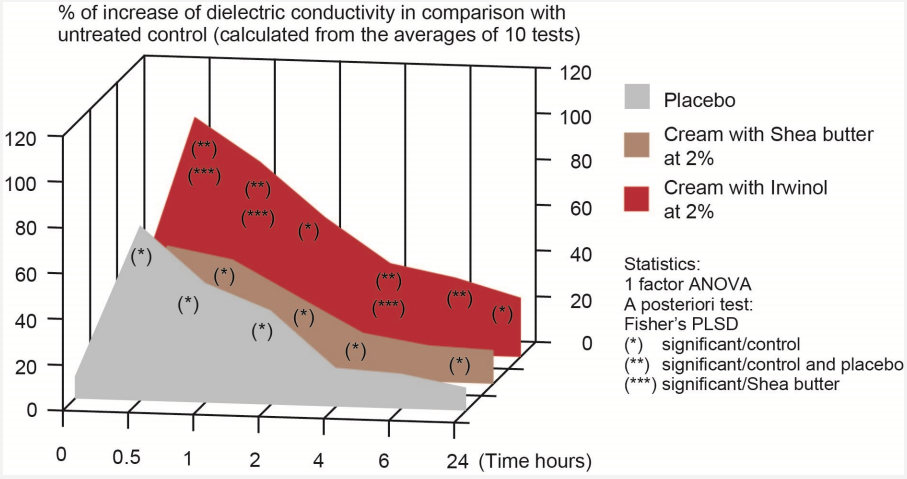


Figure 3 - Effect of Irwinol on dielectric conductivity on skin *ex vivo*.

CONCLUSION

The moisturizing effect of the cream containing 2% Irwinol is significantly stronger than that of the cream with 2%. Shea butter or the placebo cream. This effect is long-lasting, with a duration of at least 6 hours.

EFFICACY

Sensorial properties

in vivo

OBJECTIVE

The sensorial properties of Irwinol on skin were compared to those of Jojoba oil and Shea butter.

RESULTS

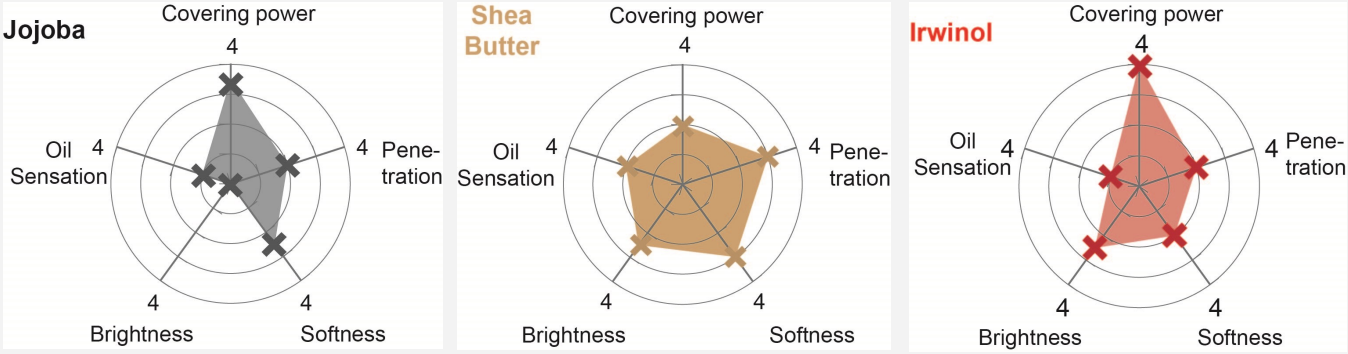


Figure 4 - Comparative sensorial profile.

CONCLUSION

Irwinol produces on the skin a soft, substantive protecting film with a bright complexion, but without any oily feeling.

EFFICACY

Softening and soothing effect *in vivo*

in vivo

OBJECTIVE

The improvement of skin softness and suppleness and the soothing effect of a cream containing Irwinol was compared to that of a placebo cream.

PROTOCOL

Ten trained judges evaluated the softness, suppleness and comfort of skin treated with a cream containing 2% Irwinol or a placebo cream. Skin softness was evaluated immediately after application. Skin suppleness and comfort were evaluated 5 min after application.

RESULTS

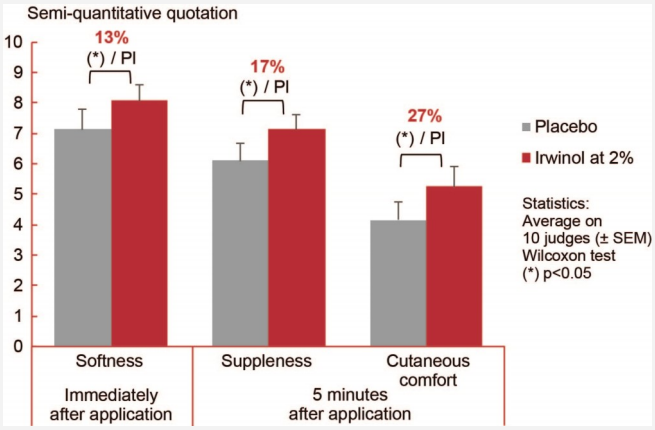


Figure 5 - Sensorial benefits of skin application of Irwinol.

CONCLUSION

The cream containing 2% Irwinol significantly improved the skin's softness (+ 13%) immediately after application and suppleness (+ 17%) and cutaneous comfort (+ 27%) 5 minutes after application compared with the placebo.

EFFICACY

Repairing activity on damaged lips *in vivo*

in vivo

OBJECTIVE

The efficacy of a lipstick containing Irwinol for repairing damaged lips was compared to that of a placebo lipstick.

PROTOCOL

Twelve volunteers with very dry lips (deep cheilitis) were treated three times a day for 7 days using a 3% Irwinol lipstick or a placebo lipstick. The appearance of their lips was documented before and after treatment, and the repairing activity was evaluated by dermatologists using a semi-quantitative scale.

RESULTS

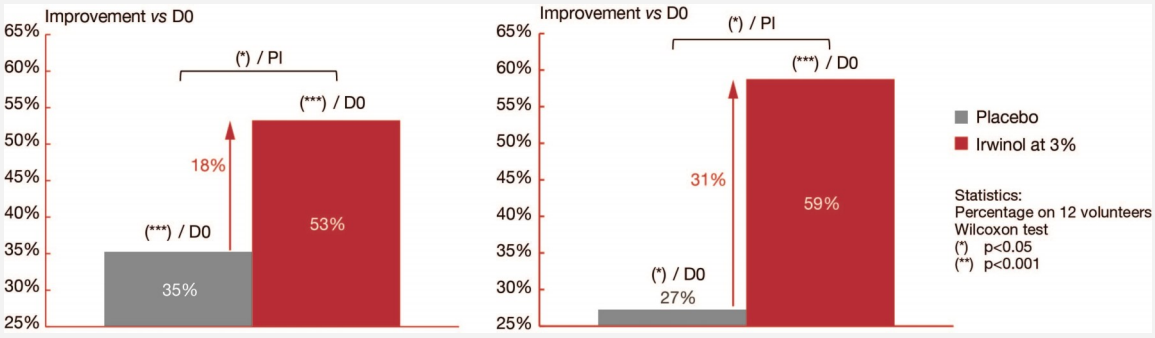


Figure 6 - Evaluation of the repairing activity.

A
Before treatment



B
After 7 days



Figure 7 - Lips **A** - Before - **B** - After treatment.

CONCLUSION

The condition of the volunteers' lips was much improved by treatment with lipsticks containing 3% Irwinol. The improvement was greater than with placebo lipsticks. Irwinol has good repairing and moisturizing activity on damaged lips.

MATERIAL&METHODS

Experimental plane
This was a double-blind randomized study, subjects are their own control.

Subject selection
12 healthy female or male volunteers aged 19 to 25 were included in the study.
All of them had marked cheilitis following treatment with retinoids and stopped any treatment on the lips 1 week before the start of the study.
2 groups of subjects were tested. Each subject has been numbered according to its order of arrival. The first group was numbered from 1 to 5; the second group according to principle 11, 21, ... 101. Only subjects with the most satisfactory criteria finally participated in the study. As a result, 12 subjects were selected from 15.

Tested products
Lipstick placebo coded 7324 B for the 1st group and 7324-1 for the second.
Lipstick with Irwinol at 3% (Batch L80337) coded 7324 A for the 1st group and 7324-2 for the second.

Treatment
Self-application by the volunteers at least three time a day, once in the morning, in the noon and in the evening, then on demand.

The subjects used each of the lip sticks for 7 consecutive days. Between the 2 weeks of treatment, during the "wash-out" period, they stopped any application on the lips for a week.

Peer subjects applied the lipstick with Irwinol at 3% the 1st week of treatment (Day 7) and the lipstick placebo the 2nd week (Day 15 to D21). The odd subjects reversed the weeks of application.

Study design
• Double blind test.
• Comparative study.
• Evolution of parameters was compared to the one before treatment (D0) and *vs* placebo.
• Volunteers served their own reference.
• Duration of the treatment: 7 days with check points at D0 and D7.

Protocol

	D0	D7	D15	D21
Admission (Informed consent)	☑			
Evaluation by Dermatologist	☑	☑	☑	☑
Macrophotograph	☑	☑	☑	☑
Randomized treatment Lipstick placebo Or Lipstick with Irwinol at 3%	From D0 to D7		From D15 to D21	
Wash out		From D7 to D15		

Evaluation method
Visual clinical examination by the Dermatologist
Cheilitis
The semi-quantitative scale used included 5 points from 0 to 4 according to the following description.
0 = normal lips, no signs of dryness. 1 = presence of light grooves. 2 = presence of deeper grooves and small scales with flat edges. 3 = presence of clearly visible grooves and scales with raised edges. 4 = presence of clearly visible grooves, scales with very raised edges and appearance of bleeding points.

Skin flakes
The semi-quantitative scale for size of the scales used included 4 points from 0 to 3 according to the following description.
0 = no visible dander. 1 = presence of scale with flat edges. 2 = presence of larger scales with raised edges. 3 = presence of large scales with very raised edges.
b. Illustrative photographs
Illustrative macrophotography before and after the treatment period for each product.

EFFICACY

Consumer test *in vivo*

in vivo

OBJECTIVE

A make-up and care kit consisting of a foundation, an eye primer and a mascara containing BASF active ingredients as well as a caring Lip lacquer containing Irwinol at 3% was evaluated in a clinical study on human volunteers.

PROTOCOL

The study was done on 64 women aged 30-50 years old, using make-up every day (included weekend days). The treatment was applied once a day for 14 days, under the normal condition of use. The volunteers answered 2 questionnaires: a well-being questionnaire related to the use of the complete kit and, a questionnaire evaluating the effectiveness of each product in the kit.

RESULTS

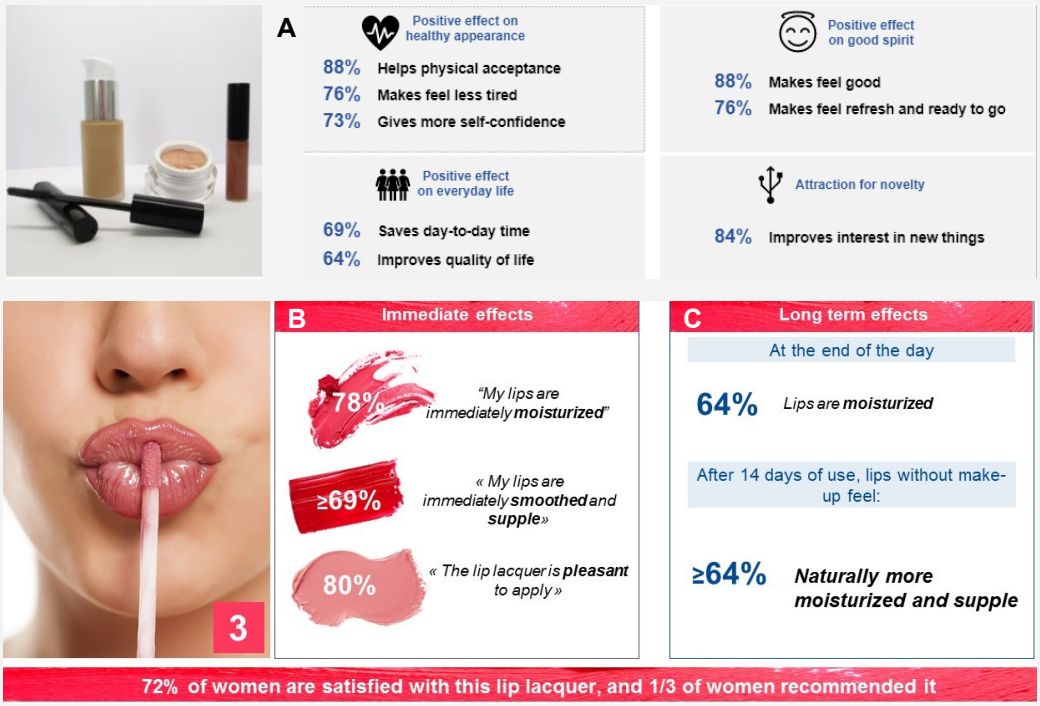


Figure 8 - Consumer study using a make-up kit (eye primer, mascara, BB cream containing BASF active ingredients ; Lip lacquer with Irwinol at 3%). **A** - Illustrative picture of make-up formula. **B** - Perceptible emotional benefits of the make-up kit among women who agreed that the make-up kit can impact the different dimensions of well-being (self-assessment, well-being questionnaire) . C - Focus on perceptible functional benefits of Irwinol in the lip lacquer (self-assessment questionnaire).

CONCLUSION

The consumer study has shown that care make-up can brings perceptible positive emotional and functional benefits. Irwinol in a lip lacquer immediately and naturally moisturizes and smoothes the lips.

ANNEXES

Annex 1 - Technical data - Available upon request

- Quality and Regulatory Product Information
- Composition sheet
- Specifications
- Formulation Data Sheet
- Halal statement
- Natural content (according ISO 16128)
- Vegan statement

Annex 2 - *In vivo* test formula

Softening and soothing effect *in vivo*- Lipstick (n°50237)

Trade Name	INCI Name	Placebo 7324A 7324-2 %	Irwinol %
Cerauba T1 (Baerlocher)	Copernicia Cerifera (Carnauba) Wax	1.86	1.86
Refined Cerilla G (Baerlocher)	Euphorbia Cerifera (Candelilla) Wax	7.54	7.54
Cerabeil White n°1 (Baerlocher)	Beeswax	0.78	0.78
Paracera M (Paramelt)	Paraffin, Microcrystalline Wax	2.45	2.45
Cerewax M85/C (Baerlocher)	Microcrystalline Wax	2.08	2.08
Finavestan A360B (Total)	Paraffinum Liquidum	15.58	15.58
Eutanol G	Octyldodecanol	14.84	14.84
Colophane Claire Type Y	Colophonium	1.89	1.89
Fina Vaseline F7850	Petrolatum	6.85	6.85
Myristyl Lactate	Myristyl Lactate	1.13	1.13
Cutina LM*	Cetearyl Alcohol, Cetearyl Glucoside, Carnauba Wax, Candelilla Wax, Beeswax, Octyldodecanol	6.57	6.57
Elestab 366*	-	q.s.	q.s.
Irwinol	Octyldodecanol, Irvingia Gabonensis Kernel Butter, Hydrogenated Coco-Glycerides	-	3.00
Castor Oil	Ricinus Communis (Castor) Seed oil	qsf.100	qsf.100

Depending on the date of the study, some ingredients may no longer be commercially available.
** BASF product no longer commercially available.*

Consumer test *in vivo*- Lip lacquer (n°10118M-B)

Trade Name	INCI Name	Irwinol %
Dehymuls PGPH	Polyglyceryl-2 Dipolyhydroxystearate	5.00
Lameform TGI	Polyglyceryl-3 Diisostearate	2.50
Dehymuls SMS	Sorbitan stearate	3.00
Cosmedia Gel CC	Dicaprylyl Carbonate, Stearalkonium Hectorite, Propylene Carbonate	13.00
Cetiol SB 45	Butyrospermum Parkii Butter	1.00
Cosmedia DC	Hydrogenated dimer dilinoleyl, dimethylcarbonate copolymer	5.00
Xiameter PMX-245	Cyclopentasiloxane	6.00
Belsil PDM 1000	Trimethylsiloxyphenyl Dimethicone	5.00
Dow Corning MQ-1640 Flake Resin	Trimethylsiloxy Silicate, Polypropyl Silsesquioxane	5.00
Cetiol Sensoft	Propylheptyl caprylate	5.00
Irwinol	Octyldodecanol, Irvingia Gabonensis Kernel Butter, Hydrogenated Coco-Glycerides	3.00
Water	-	23.80
Glycerin 99-5	Glycerin	7.00
Euxyl PE 9010	Phenoxyethanol, Ethylhexylglycerin	1.00
Sodium Chloride	Sodium Chloride	0.50
Timica Terra White	Mica, Titanium Dioxide	5.00
Timica Terra Red	Mica, Iron Oxyde	4.67
Timica Terra Black	Mica, Iron Oxyde	1.00
Timica Terra Yellow	Mica, Iron Oxyde	3.33
Covi-Ox T 70C	Tocopherol	0.10
Parfum RedLip R39403 (Technicoflor)	Fragrance	0.10

Annex 3 - Formulation examples

Caring Liquid Lip Matte - Lip lacquer (CC-FR-13-006-D29-2)

Phase	Ingredients	INCI	% by weight	Function
A	Dehymuls® PGPH	Polyglyceryl-2 Dipolyhydroxystearate	5.00	Emulsifier (W/O)
	Lameform® TGI	Polyglyceryl-3 Diisostearate	2.50	Emulsifier (W/O)
	Dehymuls® SMS	Sorbitan Stearate	3.00	Emulsifier (W/O)
	Cosmedia® Gel CC	Dicaprylyl Carbonate, Stearalkonium Hectorite, Propylene Carbonate	13.00	Rheology modifier
	Cetiol® SB 45	Butyrospermum Parkii Butter	1.00	Emollient
	Cosmedia® DC	Hydrogenated Dimer Dilinoleyl/Dimethylcarbonate Copolymer	5.00	Film forming agent
	Belsil PDM 1000 *(WACKER)*	Trimethylsiloxyphenyl Dimethicone	5.00	Emollient
	Dow Corning MQ-1640 Flake Resin *(Dow Corning)*	Trimethylsiloxy Silicate, Polypropyl Silsesquioxane	5.00	Film forming agent
	Cetiol® Sensoft	Propylheptyl Caprylate	5.00	Emollient
	Xiameter PMX-0245 Cyclopentasiloxane *(Dow Corning)*	Cyclopentasiloxane	6.00	Emollient
	Irwinol™ LS 9890	Octyldodecanol, Irvingia Gabonensis, Hydrogenated Coco-Glycerides	3.00	Active ingredient
B	Water, demin.	Aqua	23.70	
	Glycerin	Glycerin	7.00	Humectant
	Sodium Chloride	Sodium Chloride	0.50	Stabilizer
	Euxyl PE 9010 *(Schülke)*	Phenoxyethanol, Ethylhexylglycerin	1.10	Preservative
C	Timica® Terra White MN4501	Mica, Titanium Dioxide	5.00	Effect pigment
	Timica® Terra Red MN4506	Mica, Iron Oxides, Titanium Dioxide	4.67	Effect pigment
	Timica® Terra Black MN4498	Mica, Iron Oxides, Titanium Dioxide	1.00	Effect pigment
	Timica® Terra Yellow MN4502	Mica, Iron Oxides, Titanium Dioxide	3.33	Effect pigment
D	Covi-ox® T 70 C	Tocopherol	0.10	Antioxidant
	Perfume*	Parfum	0.10	Fragrance

Specifications	
Viscosity	30000-35000 mPa s
(Brookfield; RVT; spindle TC, Helipath; 20 rpm; 20°C)	

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